

PBL型演習

GROUP C

Learning-Based Temporal Motion Prediction for Risk- Aware Autonomous Driving

学習ベースの時間的歩行者動作予測によるリスク認識型自動運転支援



Presented By:

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- Benjamin Jung



Objective

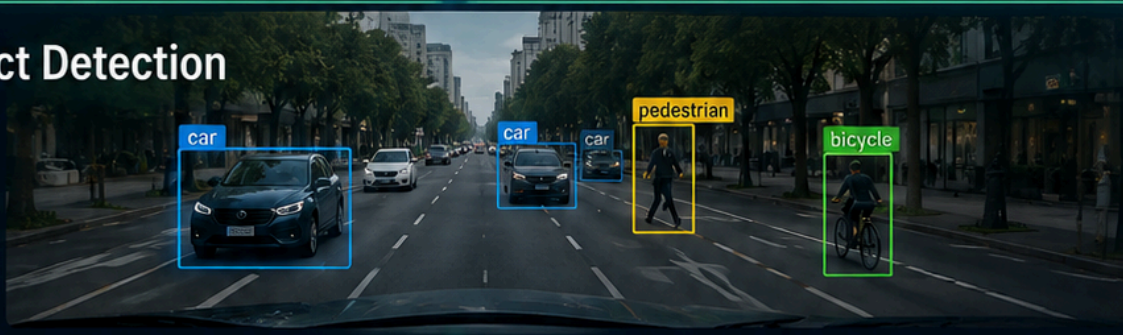
- Develop a learning-based temporal pedestrian motion prediction system for autonomous driving.
- Extract spatial features from RGB video sequences using ResNet-18.
- Learn temporal motion patterns using LSTM and BiLSTM networks.
- Predict future pedestrian motion to enable earlier risk assessment.
- Compare the performance of CNN→LSTM and CNN→BiLSTM architectures in MATLAB.
- Evaluate the proposed approach using the argoverse Tracking dataset.

PIPELINE

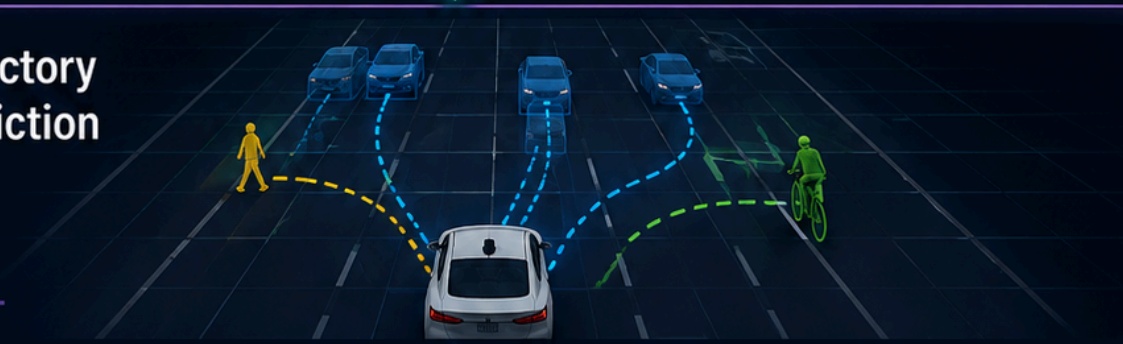
1 Camera Images



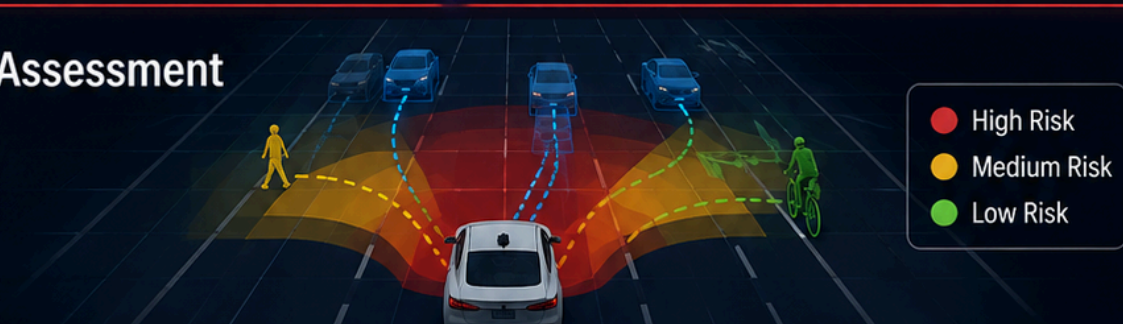
2 Object Detection



3 Trajectory Prediction



4 Risk Assessment



Learning-Based Temporal Motion Prediction for
Risk-Aware Autonomous Driving

Project Motivation

Why is this project important?

Current ADAS

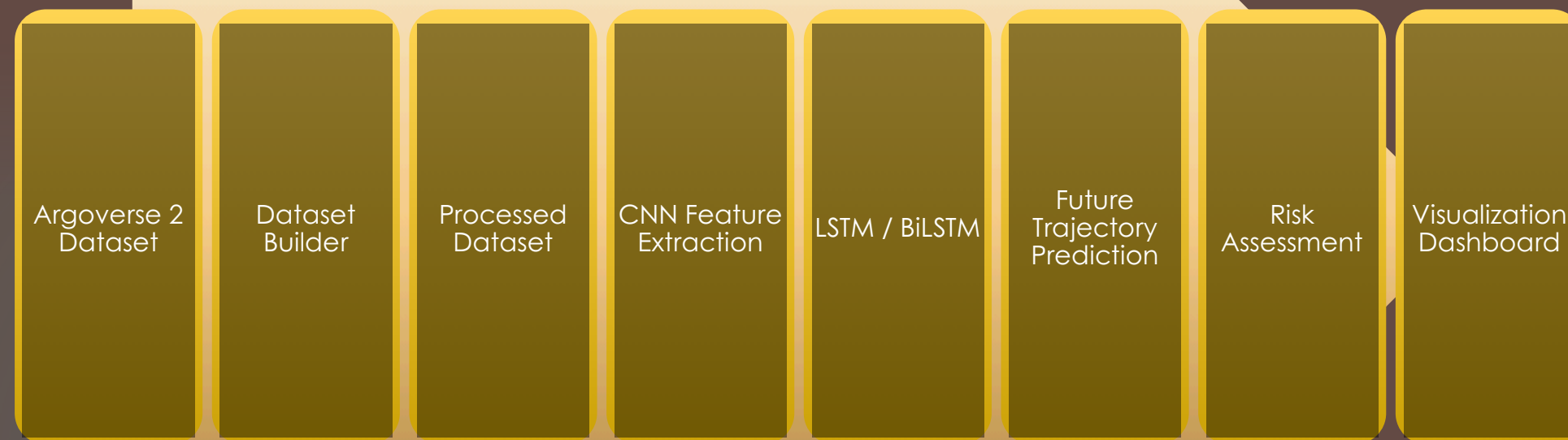
- ✓ Detects vehicles and pedestrians
- ✓ Understands the current traffic scene
- ✓ Cannot predict future movements

Our Goal

- ✓ Develop a system that can
- ✓ Predict future pedestrian trajectories
- ✓ Estimate potential collision risk
- ✓ Improve autonomous driving safety

Project Overview

Proposed System



Key Features

- Modular architecture
- Temporal motion prediction
- Automatic risk estimation
- End-to-end processing pipeline

学習ベースの時間的歩行者動作予測によるリスク認識型自動運転支援

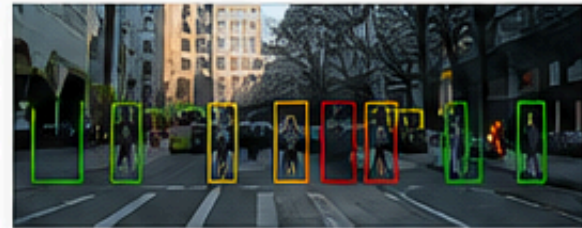
～ RGBビデオと深層学習による将来の歩行者軌跡予測とリスク評価を実現 ～

センサ（入力）



データセット

Argoverse 2 Sensor Dataset
(高精度なトラッキングアノテーション)



データの特徴

- ✓ 歩行者・車両の長期トラッキング
- ✓ 3Dワールド座標・速度情報
- ✓ 高品質な時系列アノテーション
- ✓ 大規模シーンでの多様な交通環境

データ分割（オブジェクト数）

Train	約 292,000
Validation	約 53,000
Test	約 57,000

入力（Input）



過去10フレームのRGBビデオシーケンス
(10フレーム = 約2秒)



空間特徴抽出（Feature Extraction）

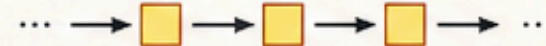


ResNet-18
(ImageNet事前学習)

出力次元: 512

時間的特徴学習（Temporal Modeling）

モデル A : CNN → LSTM



モデル B : CNN → BiLSTM



将来軌跡予測（Future Trajectory Prediction）



現在位置

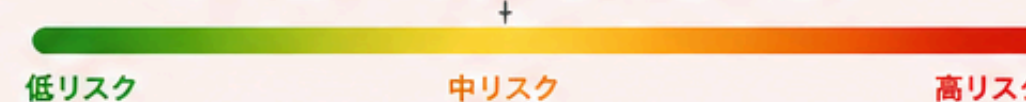
5ステップ未来予測



リスク評価（Risk Assessment）



TTC（衝突までの時間） + 予測軌跡の不確実性



リスク認識型自動運転支援（出力）

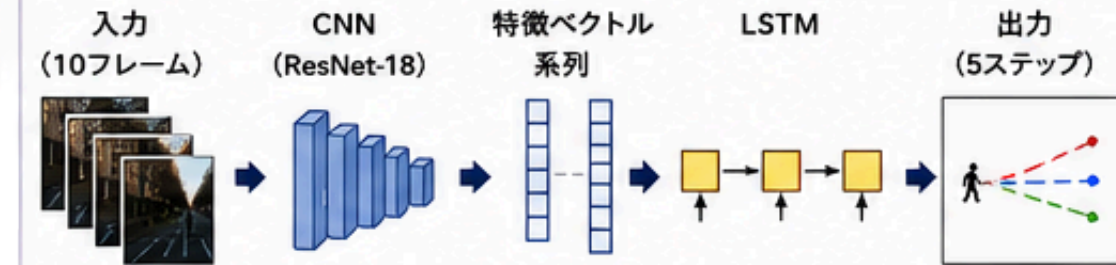


早期警告・危険度通知
(ドライバ / 自動操舵システムへ通知)

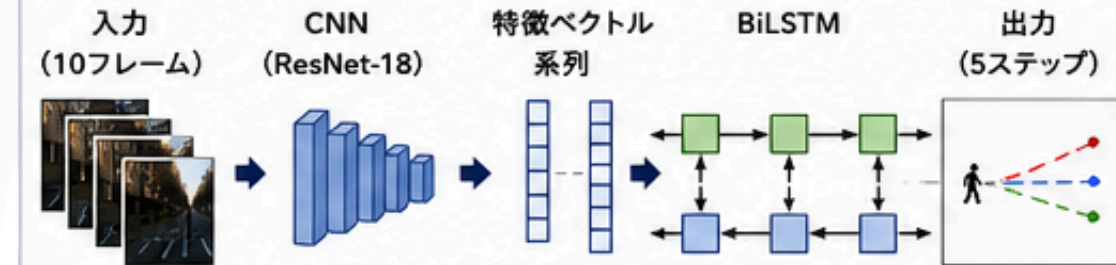


候補ネットワーク（Candidate Networks）

アーキテクチャ A : CNN → LSTM



アーキテクチャ B : CNN → BiLSTM



モデル比較（LSTM vs BiLSTM）

比較項目	CNN → LSTM	CNN → BiLSTM	備考
時間情報の利用	過去 → 未来（片方向）	過去 ⇄ 未来（双方向）	BiLSTMは将来からの文脈も考慮
文脈の理解	限定的	より豊か	長期依存関係の学習に有利
予測精度（傾向）	良い	やや良い	ADEでわずかに優位
計算コスト	低い（速い）	高い（やや遅い）	双方向のため計算量増

本システムの特徴

- 深層学習による時間的歩行者軌跡予測
- 将来のリスクを定量的に評価（リスクスコア・TTC）
- モジュール化されたエンドツーエンドパイプライン
- 可視化ダッシュボードによる直感的な結果提示

凡例



カメラ入力



CNN特徴抽出



LSTM



BiLSTM

未来予測ステップ



リスク評価・警告

Dataset Preparation

Dataset Selection

Selected Dataset

Argoverse 2 Sensor Dataset

Reasons:

- RGB camera images
- Object tracking annotations
- World coordinate information
- High-quality labels
- Suitable for trajectory prediction

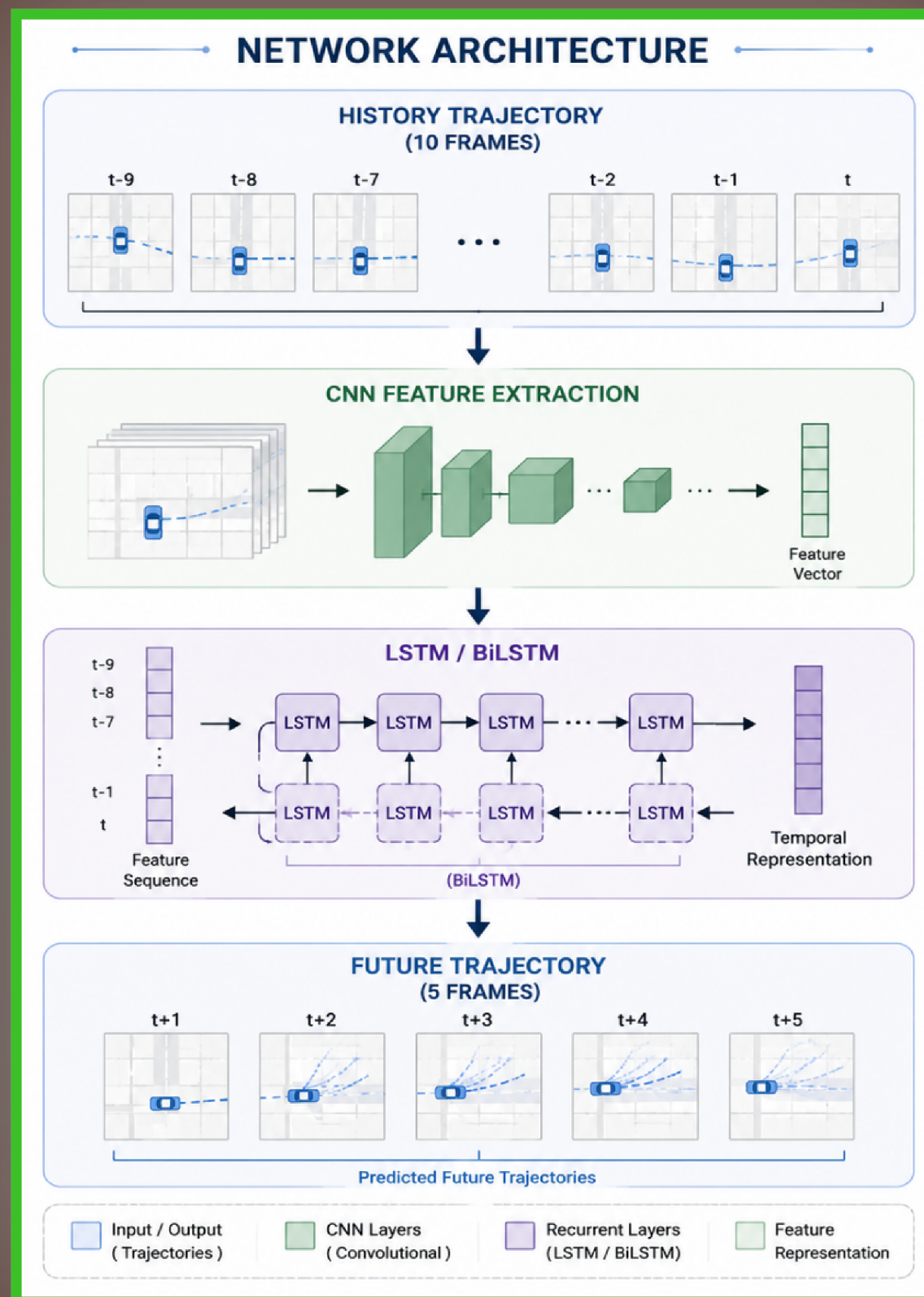
Dataset Builder

Implemented:

- Dynamic object filtering
- History trajectory extraction
- Future trajectory extraction
- Train / Validation / Test split
- JSON dataset generation

Dataset	Samples
Train	~292,000
Validation	~53,000
Test	~57,000

Temporal Prediction Model



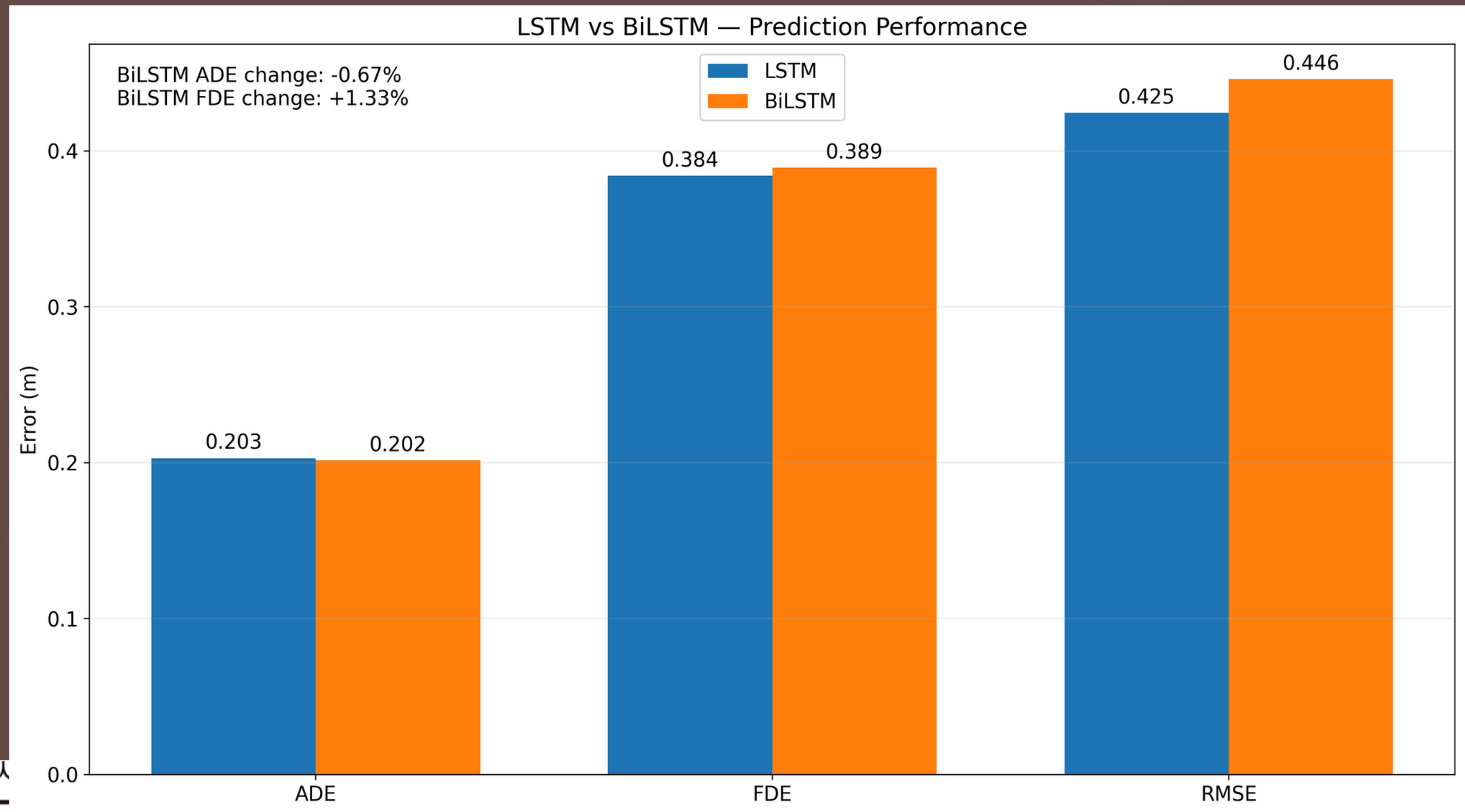
Features

- Predicts future pedestrian motion
- Supports both LSTM and BiLSTM
- Model selection through configuration
- Flexible and scalable design

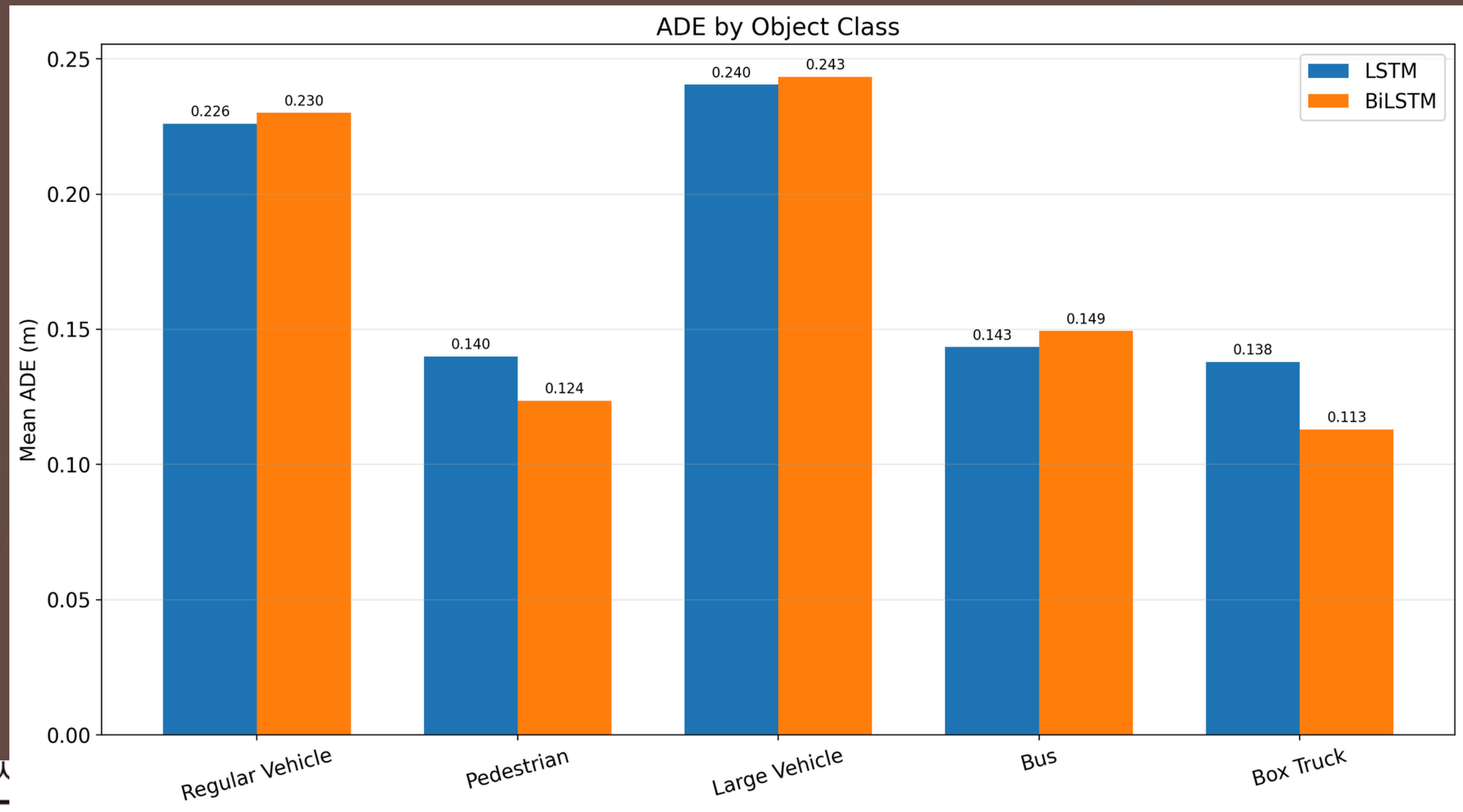
finished Progress

Module	Status
Project Definition	✓
Dataset Selection	✓
Dataset Builder	✓
Temporal Prediction Model	✓
Training Pipeline	✓
Evaluation Module	✓
Risk Assessment	✓
Inference Pipeline	✓
Visualization Dashboard	✓

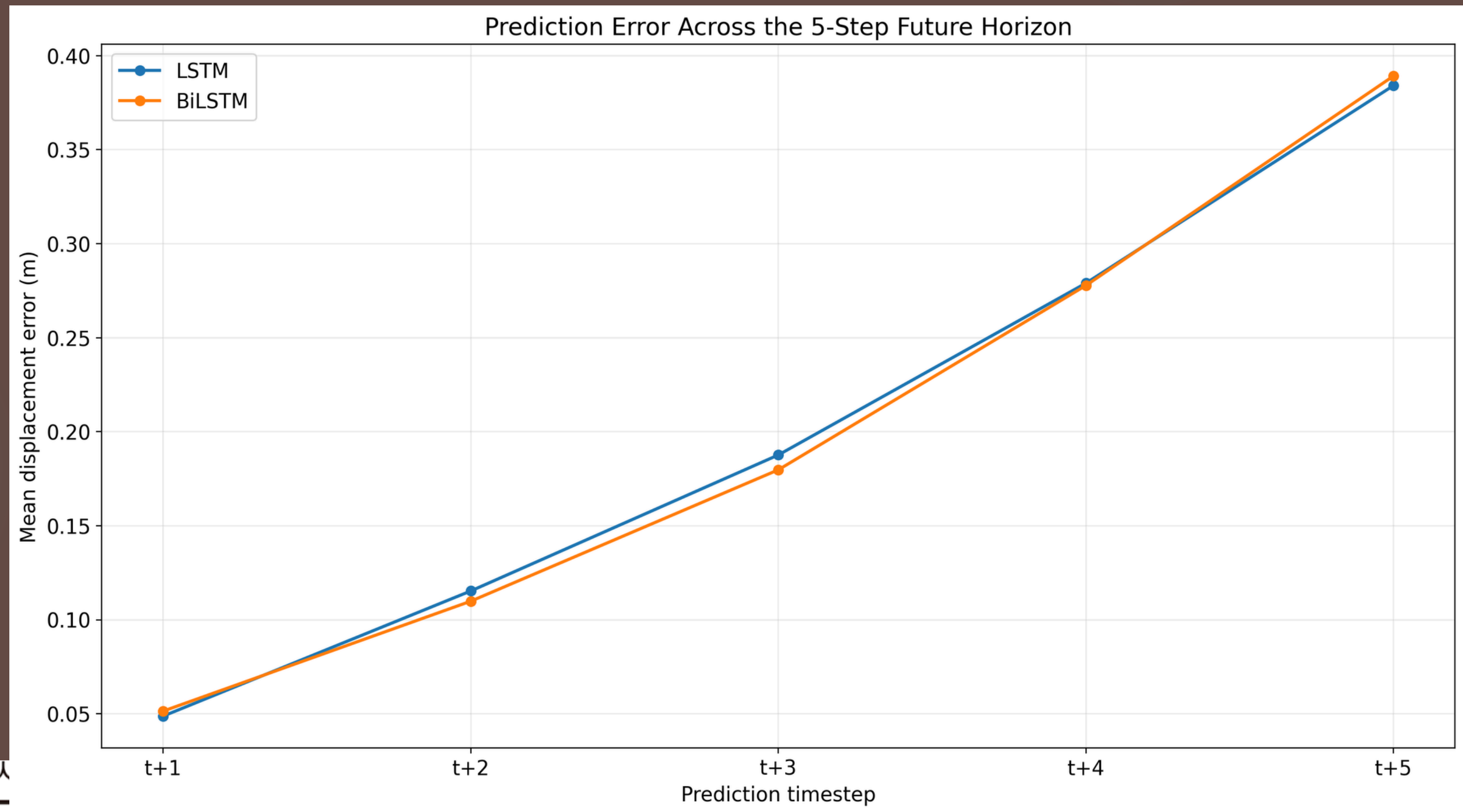
RESULTS



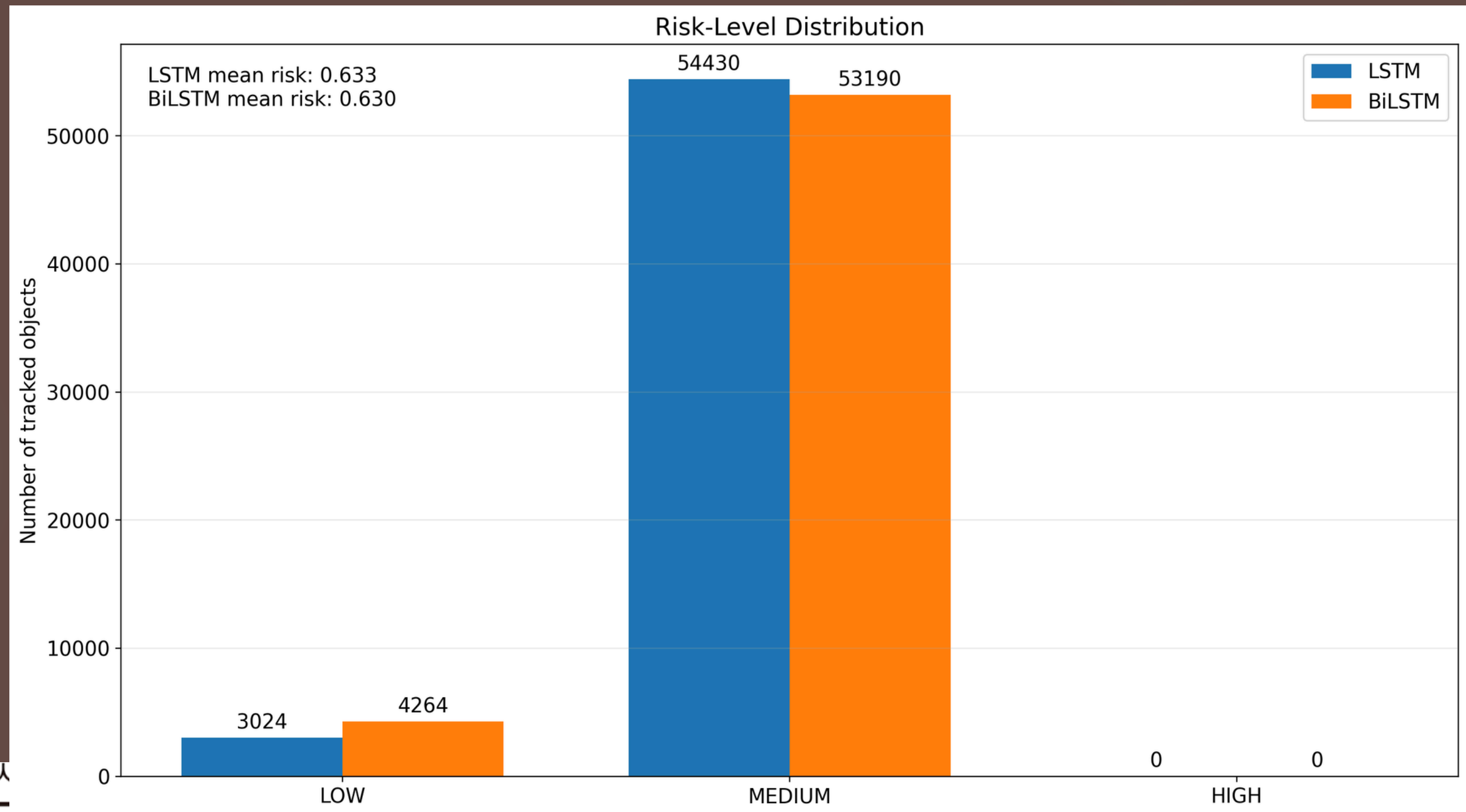
RESULTS



RESULTS



RESULTS



RESULTS

Quantitative Comparison

Metric	LSTM	BiLSTM
Mean ADE (m)	0.203	0.202
Mean FDE (m)	0.384	0.389
RMSE (m)	0.425	0.446
Median ADE (m)	0.112	0.102
Median FDE (m)	0.204	0.185
Mean Risk Score	0.633	0.630

Learning-Based Temporal Motion Prediction for Risk-Aware Autonomous Driving

RGB camera sequence | BiLSTM | Argoverse 2 Sensor Dataset

Jayanth Narayana K
FRAME 000160

● Camera —● History —● LSTM / BiLSTM —● Future Prediction —● ADE / FDE —● Risk

CAMERA VIEW

Current RGB frame
2D boxes unavailable in results.json



BIRD'S-EYE VIEW

AV2 ego-reference coordinates | current object = anchor

2 nearby / 47 total tracks



FRAME SUMMARY

TRACKS	47	AVG RISK	0.640
AVG ADE	0.105 m	LOW	0
AVG FDE	0.201 m	MEDIUM	47
		HIGH	0

History: 10 frames
Prediction horizon: 5 future points

PRIORITY OBJECTS

V3330 REGULAR_VEHICLE MEDIUM Risk: 0.65 ADE: 0.245 m FDE: 0.508 m TTC: 0.99 s Speed: 1.14 m/s	V3142 REGULAR_VEHICLE MEDIUM Risk: 0.65 ADE: 0.397 m FDE: 0.783 m TTC: 0.99 s Speed: 0.88 m/s	V3180 REGULAR_VEHICLE MEDIUM Risk: 0.64 ADE: 0.198 m FDE: 0.406 m TTC: 0.98 s Speed: 0.54 m/s	V3117 REGULAR_VEHICLE MEDIUM Risk: 0.64 ADE: 0.111 m FDE: 0.246 m TTC: 1.81 s Speed: 0.52 m/s
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— History — Prediction — Ground Truth — Risk

Relative trajectories + anchor | Prediction vs GT | Risk-aware evaluation

Learning-Based Temporal Motion Prediction for Risk-Aware Autonomous Driving

RGB camera sequence | BiLSTM | Argoverse 2 Sensor Dataset

Jayanth Narayana K
FRAME 000200

● Camera —● History —● BiLSTM —● Future Prediction —● ADE / FDE —● Risk

CAMERA VIEW

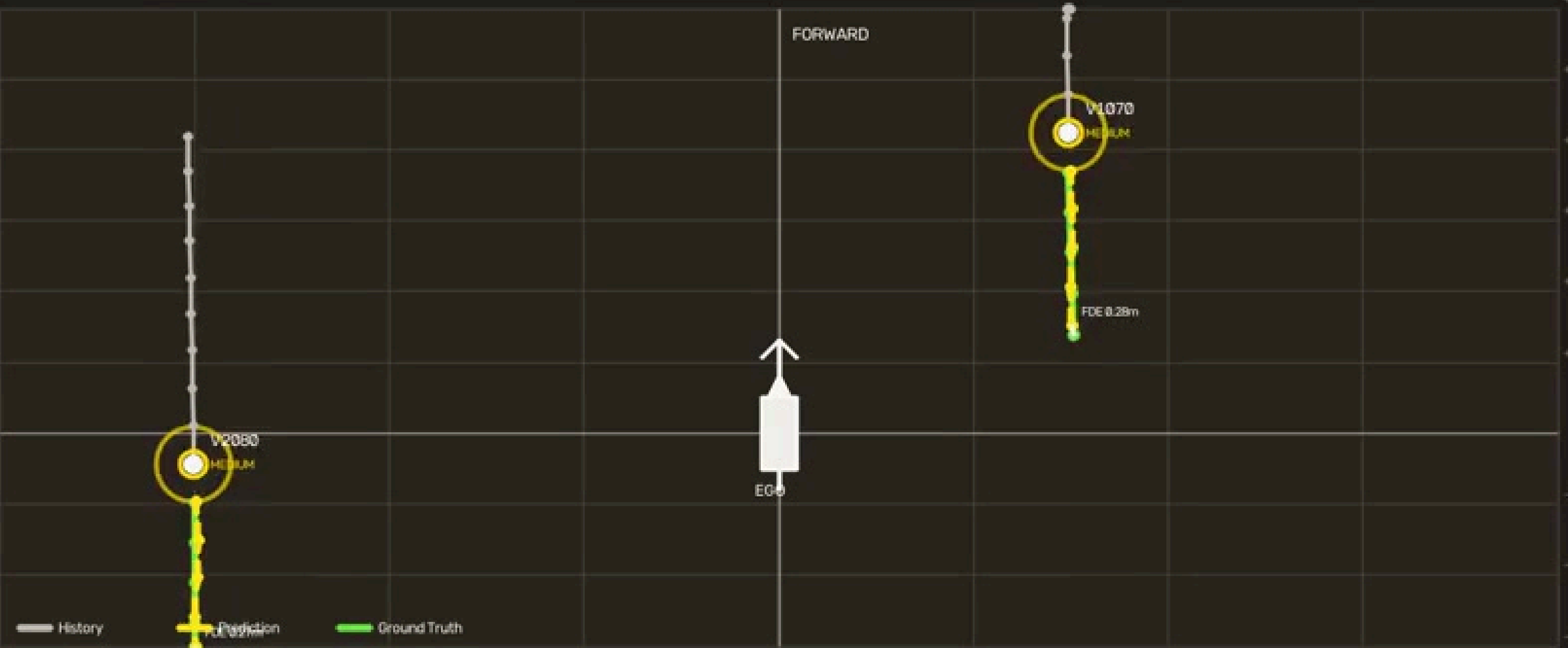
Current RGB frame
2D boxes unavailable in results json



BIRD'S-EYE VIEW

AV2 ego-reference coordinates | current object = anchor

2 nearby / 29 total tracks



FRAME SUMMARY

TRACKS	29	AVG RISK	0.648
AVG ADE	0.171 m	LOW	0
AVG FDE	0.301 m	MEDIUM	29
		HIGH	0

History: 10 frames
Prediction horizon: 5 future points

PRIORITY OBJECTS

V3370 REGULAR_VEHICLE MEDIUM Risk 0.65 ADE 0.086 m FDE 0.123 m TTC 0.99 s Speed 1.97 m/s	V3017 REGULAR_VEHICLE MEDIUM Risk 0.65 ADE 0.206 m FDE 0.379 m TTC 0.98 s Speed 1.11 m/s	V3519 REGULAR_VEHICLE MEDIUM Risk 0.65 ADE 0.211 m FDE 0.388 m TTC 0.98 s Speed 1.11 m/s	V3957 REGULAR_VEHICLE MEDIUM Risk 0.65 ADE 0.220 m FDE 0.402 m TTC 0.98 s Speed 1.11 m/s
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— History — Prediction — Ground Truth — Risk

Relative trajectories + anchor | Prediction vs GT | Risk-aware evaluation

Conclusion

- Developed an RGB-based temporal motion prediction pipeline for risk-aware autonomous driving.
- Used ResNet-18 + LSTM/BiLSTM to predict 5 future trajectory steps from 10-frame motion history.
- Achieved approximately 0.20 m mean ADE on the evaluated results.
- LSTM and BiLSTM showed comparable prediction performance, with different strengths across ADE, FDE and RMSE.
- Integrated trajectory prediction with risk score and TTC-based risk assessment.
- Developed an integrated visualization combining camera view, trajectory prediction, ground truth, BEV and risk information.
- Demonstrated how future motion prediction can provide information for earlier risk awareness.

Future Work

- The system provides a strong foundation for temporal motion prediction, with ongoing improvements in model accuracy and performance.
- Future work focuses on advanced experiments and enhancing the visualization dashboard for a clearer final demonstration.

Ongoing Development

- **Improve prediction accuracy**
- **Test more real-world driving scenarios**
- **Improve risk assessment**
- **Real-time system implementation**

4

Work Plan & Task Assignment

作業計画・担当分担 (Group C)

Project Timeline (作業計画)

Period	Activities
~ Jul. 17	Literature review Project definition Dataset selection (KITTI Tracking) Network architecture design
~ Jul. 24	VScode environment setup Data preprocessing ResNet-18 feature extraction
~ Jul. 31	CNN→LSTM implementation CNN→BILSTM implementation Training & Validation
~ Aug. 18	Performance comparison Presentation preparation Final report

Task Assignment (担当分担)

Student	ID	Responsibility
Jayanth K	266E0218	Literature review, CNN→LSTM Implementation, system integration
Ragav	266E0228	CNN→BILSTM Implementation, model training & evaluation
Benjamin	26600601	argoverse dataset preparation, preprocessing, visualization, documentation

Group Activities

- Weekly meeting (2 hour)
- Progress discussion
- Final presentation preparation
- Final report writing



ありがとうございました

感謝

感謝の心を忘れずに

和

ご清聴ありがとうございました。

Thank You for Your Attention

SHANKAR RAGAVENDAR

JAYANTH

BENJAMIN